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Can Environmental Policy Encourage Technical Change? Emissions Taxes and R&D Investment in Polluting Firms

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1 Big picture

- **Question.** Do market-based environmental policies induce dirty firms to make the technology investments needed to clean up production, or do they mainly affect patentable “green” invention at the product level?
- **Core idea.** Taxes on dirty emissions make incumbent production technologies more expensive to operate. Polluting firms should then invest more in the technological capabilities required to adopt cleaner production processes.
- **Setting.** Non-U.S. manufacturing firms from 18 OECD countries, observed from 1990 to 2012, merged with country-year taxes on sulfur oxide (SO_x) emissions.
- **Main empirical object.** Country-level variation in SO_x taxes, combined with cross-industry differences in pollution intensity measured from U.S. industry SO_x emissions data.
- **Main result.** Higher SO_x taxes are associated with higher firm-level R&D spending, and the response is concentrated in the dirtier industries that are most exposed to the tax.
- **Magnitude.** The baseline coefficient on pollution taxes is about 0.06 in logs. The differential effect for firms in above-median pollution industries is about 0.08–0.10, implying an extra R&D response of roughly 14%–18% of average R&D for a one-standard-deviation increase in the tax.
- **Key twist.** Polluting firms increase R&D, but usually do *not* increase overall patenting. This suggests the response is not mainly “first-face” R&D aimed at patentable invention.
- **Mechanism.** The paper argues that pollution taxes increase the value of the “second face” of R&D: investment in absorptive capacity, process improvement, and the ability to use external knowledge and know-how.
- **Supporting evidence.** The R&D response is strongest in low-invention sectors, high-process-R&D sectors, and high-spillover / low-appropriability sectors. Pollution taxes also raise the market value of R&D in precisely these subsets.
- **Why it matters.** The paper broadens the usual innovation-and-environment story. Environmental policy may shift technical change not only by stimulating clean patents, but also by inducing dirty firms to build the technological capacity needed to re-engineer production.

2 Data and measurement (key objects)

2.1 Firm-level sample

- **Source.** Compustat Global plus Canadian firms from Compustat North America.
- **Sample restrictions.** Non-U.S. firms, fully consolidated statements, manufacturing SIC 2000–3999, and at least three nonmissing R&D observations during 1990–2012.
- **Coverage.** Roughly 33,500 firm-years across 18 countries over 1990–2012. In the patent-matched sample, the paper reports 3,594 unique firms.
- **Why this restriction matters.** The authors need repeated within-firm observations on R&D to identify a response to tax changes, and enough within-country industry variation to compare dirtier and cleaner firms.

2.2 Pollution-tax measure

- **Source.** OECD Environmental Stringency Index and the Policy Instruments for the Environment (PINE) database.
- **Variable.** $Pollution\ taxes_{c,t}$ is a categorical measure, from 0 to 6, of taxes and charges directly applied to SO_x emissions in country c and year t .
- **Economic meaning.** Higher values indicate more stringent taxes on dirty manufacturing emissions.
- **Why SO_x?** The paper emphasizes that SO_x is a major air pollutant and that there is substantial variation across countries and over time in SO_x taxes.

2.3 Pollution intensity / treatment exposure

- **Source.** Levinson (2009), using pounds of SO_x emissions per unit of output in U.S. manufacturing industries in 1987.
- **Unit.** Three-digit SIC industry.
- **Purpose.** This provides the cross-sectional exposure measure: dirtier industries should be more heavily treated by higher pollution taxes.
- **Descriptive spread.** Hydraulic cement (SIC 324) is the most SO_x-intensive industry in the sample with emissions around 140 pounds per unit of output; the 10 least-polluting industries are near zero.
- **Identification logic.** The authors do not need pollution intensity to be identical across countries; they need the relative ordering of industries by dirtiness to be broadly stable across countries.

2.4 Innovation and patent data

- **Source.** The European Patent Office's Worldwide Patent Statistical Database (PATSTAT).
- **Main patent measure.** New triadic patents, dated by application year.
- **Alternative output measure.** Future citations to granted patents.
- **Clean invention measure.** Patents in air-pollution-abatement technology classes using the OECD's ENV-TECH taxonomy.
- **Interpretation.** These data let the paper distinguish between more R&D spending and more patentable innovation.

2.5 Market-value data

- **Source.** Compustat Global Security Daily.
- **Use.** To test whether the market values a marginal dollar of R&D more when pollution taxes are higher.
- **Why important?** Since polluting firms often increase R&D without increasing patent counts, this market-value test is crucial to show that the extra R&D is not simply wasteful or spurious.

2.6 Main constructed variables

- **$R\&D_{i,t}$.** Log R&D expenditure of firm i in year t .
- **$Sales_{i,t}$.** Log firm sales, used as an output scale control.
- **$Patent\ count_{i,t}$.** Log of one plus triadic patent counts.
- **$Stock\ of\ patents_{i,t-1}$.** Patent stock used to control for preexisting innovative capacity.
- **$Pollution\ tax\ change$.** Event-style indicator that separates years before, at, and after tax increases.
- **Quartiles of SOxpolluters.** Indicators for industry pollution-intensity quartiles, used in the differential-exposure design.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline R&D specification

The central estimating equation is

$$R\&D_{i,t} = \beta\ PollutionTaxes_{c,t-1} + \gamma\ Sales_{i,t} + \eta_i + \eta_t + \varepsilon_{i,t}. \quad (1)$$

- **Dependent variable.** Log R&D spending of firm i .
- **Regressor of interest.** The lagged country-level SOxtax.
- **Controls.** Firm fixed effects, year fixed effects, and in some specifications cash flow/assets, sales growth, cash holdings/assets, debt/assets, and an IFRS indicator.
- **Interpretation.** $\beta > 0$ means that stricter emissions taxes are associated with more technology investment by firms.
- **Main estimate.** The baseline $\hat{\beta}$ is about 0.062 without controls and 0.055 with controls.

3.2 Event-timing decomposition

To verify timing and rule out pretrends, the paper decomposes pollution-tax changes into leads, contemporaneous changes, and post-change periods:

$$R\&D_{i,t} = \beta_{-1}TaxChange_{c,t}^{(-1)} + \beta_0TaxChange_{c,t}^{(0)} + \beta_{\geq 1}TaxChange_{c,t}^{(\geq 1)} + \gamma\ Sales_{i,t} + \eta_i + \eta_t + \varepsilon_{i,t}. \quad (2)$$

- **Purpose.** This checks whether R&D starts rising *before* the tax change, which would weaken a causal interpretation.
- **Main result.** The lead coefficient is essentially zero; the effect shows up only after tax increases. The post-change coefficient is about 0.136–0.156.
- **Interpretation.** There is no evidence that R&D is already trending up before taxes increase.

3.3 Differential-exposure specification

The core causal design augments the baseline regression with interactions between pollution taxes and industry dirtiness:

$$R\&D_{i,t} = \beta_0 PollutionTaxes_{c,t-1} + \sum_{k=2}^4 \beta_k PollutionTaxes_{c,t-1} \times 1\{Q_i^{SOX} = k\} + \gamma Sales_{i,t} + \eta_i + \eta_{c,t} + \eta_{j,t} + \varepsilon_{i,t}, \quad (3)$$

where the fixed effects vary by specification.

- **Idea.** Firms in dirtier industries are more exposed to emissions taxes than firms in cleaner industries.
- **Interpretation.** Positive β_k for the dirtier quartiles means that higher taxes have stronger effects where the tax bites more.
- **Main result.** The interaction coefficients for the top two pollution quartiles are around 0.08–0.10 and remain significant with country-year and even industry-year fixed effects.
- **Economic magnitude.** A one-standard-deviation increase in pollution taxes raises R&D by about 0.15–0.19 more in above-median pollution industries than in the cleanest quartile.

3.4 Innovation-output specification

To test whether pollution taxes increase patentable innovation, the paper estimates variants of the baseline equation with patents as the dependent variable:

$$PatentCount_{i,t} = \beta PollutionTaxes_{c,t-1} + \rho PatentStock_{i,t-1} + \gamma Sales_{i,t} + \eta_i + \eta_t + \varepsilon_{i,t}. \quad (4)$$

- **All patents.** There is no meaningful effect of pollution taxes on total triadic patenting.
- **Clean patents.** Pollution taxes are positively associated with air-pollution-abatement patents in the full sample, but the high-pollution firms that drive the R&D response do not disproportionately increase patenting.
- **Interpretation.** The extra R&D is usually not the “first face” of R&D aimed at new patentable products.

3.5 Market-value specification

The paper’s second numbered equation studies whether higher taxes change how the market values R&D:

$$\frac{MV_{i,t}}{MV_{i,t-1}} = \alpha_1 \frac{R\&D_{i,t}}{MV_{i,t-1}} + \alpha_2 \left(PollutionTaxes_{c,t} \times \frac{R\&D_{i,t}}{MV_{i,t-1}} \right) + X_{i,t} + \eta_i + \eta_{c,t} + \varepsilon_{i,t}. \quad (5)$$

- **Dependent variable.** The percentage change in market value of equity.
- **Key parameter.** α_2 captures whether a marginal dollar of R&D becomes more valuable when pollution taxes are higher.

- **Controls.** Net assets, change in net income, dividends, cash holdings, net financing, and debt, all scaled by lagged market value.
- **Main result.** In above-median pollution industries, the interaction is strongly positive (1.431 or 0.685, depending on controls). In below-median pollution industries, it is negative and insignificant.
- **Interpretation.** The market values R&D more highly exactly where pollution taxes should increase the payoff to process-improving technological capability.

3.6 Mechanism subsample specifications

The paper then re-estimates Equation (1) on targeted subsamples:

- **Low-invention sectors.** Sectors with below-median patent stocks, or sectors outside the small set that historically generate most air-pollution-abatement patents.
- **High process-R&D sectors.** Industries where process R&D is a larger share of total R&D, using Cohen and Klepper (1996).
- **High-spillover sectors.** Industries with low appropriability / high knowledge spillovers using Cohen, Nelson, and Walsh (2000).
- **Interpretation.** If pollution taxes mainly stimulate absorptive-capacity investment rather than invention, the R&D response should be concentrated in these sectors.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Not a classical IV paper.** The main design is a differential-exposure strategy rather than an external instrument.
- **Source of variation.** The paper uses within-country changes in SO_x taxes over time, plus cross-industry differences in pollution intensity.
- **Comparison.** Dirty industries in tax-increasing countries are compared to cleaner industries in the same countries and years, and to dirty industries in countries without similar tax changes.
- **Rajan-Zingales logic.** Preexisting industry exposure determines which firms are more “treated” by a common country-level policy change.

4.2 Why the estimates can be read causally

- **Timing.** R&D does not rise prior to tax increases; it rises only after taxes change.
- **Exposure gradient.** The tax effect is systematically larger in the top pollution quartiles.
- **Fixed effects.** Firm fixed effects absorb time-invariant firm and country differences; country-year fixed effects absorb time-varying country shocks; industry-year fixed effects absorb time-varying industry shocks.

- **Market-value corroboration.** The taxes raise the value of R&D in the most exposed firms, not just the level of R&D.

4.3 Main threats and responses

- **Alternative country-level mechanisms.** The paper checks whether development (GDP per capita), public environmental R&D, or the user cost of R&D can explain the results. They do not.
- **Common boom in 1997–2001.** The paper interacts a boom-period dummy with pollution-intensity indicators and shows that the results do not reflect a broad global upswing hitting dirty industries at the same time as taxes rise.
- **Accounting differences.** The paper studies IFRS adoption at both the country and firm level and finds no evidence that accounting changes explain the R&D response.
- **Industry growth opportunities.** Results survive controls for industry shares of country R&D and employment.
- **Country composition.** Results are not driven by Japan, Canada, or the U.K., and they survive dropping countries with no tax variation.

4.4 Key conceptual identification distinction

- **Clean patents versus absorptive capacity.** A rise in clean patents would point to the first face of R&D; a rise in R&D without patents, concentrated in high-spillover / low-appropriability sectors, points to the second face.
- **What the paper identifies.** The evidence supports the view that emissions taxes induce dirty firms to invest in process-improving, knowledge-absorbing capability rather than only in patentable invention.

5 Tables and figures

5.1 Figure 1: Taxes on SO_x emissions, 1990–2012

Read:

- The figure plots the time path of country-level SO_x taxes across the eight countries in the sample that ever tax SO_x emissions.
- The key message is that there is meaningful within-country time-series variation, not just static cross-country differences.
- Japan has a persistently high tax level; Denmark and Korea exhibit especially large changes; several countries never tax SO_x and act as controls.

5.2 Table 1: Description of variables

Read:

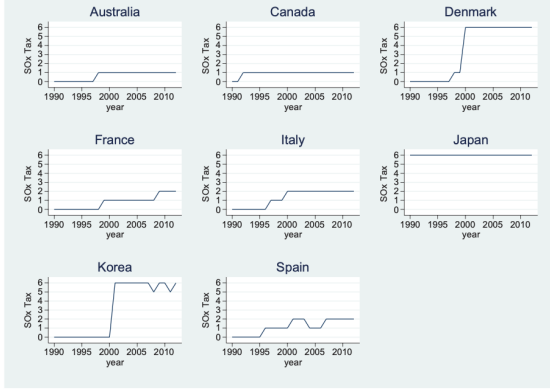


Figure 1
Taxes on SO_x emissions, 1990 to 2012
This figure reports the evolution of taxation on sulphur oxides (SO_x) by country and over time. Our sample also includes 10 countries with no SO_x tax: Austria, Belgium, Finland, Germany, Greece, Ireland, Netherlands, Norway, Sweden, and the United Kingdom. Source: OECD.

Table 1
Description of variables

Variable name	Definition	Source
$R\&D_{i,t}$	The natural logarithm of firm-level research and development (R&D) expenditures (deflated to US\$(2000) millions and winsorized at the 1% level)	Compustat Global and North America
$Sales_{i,t}$	The natural logarithm of firm level sales (deflated to US\$(2000) millions and winsorized at the 1% level)	Compustat Global and North America
$Cash-flow-to-assets_{i,t}$	Cash flow divided by the beginning of year book value of total assets (winsorized at the 1% level)	Compustat Global and North America
$Sales\ growth_{i,t}$	First difference in Sales (winsorized at the 1% level)	Compustat Global and North America
$Cash\ holdings-to-assets_{i,t}$	Cash holdings divided by the beginning of year book value of total assets (winsorized at the 1% level)	Compustat Global and North America
$Total\ debt-to-assets_{i,t}$	Total debt divided by the book value of total assets (winsorized at the 1% level)	Compustat Global and North America
$IFRS_{i,t}$	An indicator variable taking on the value one if the firm reports using International Financial Reporting Standards (IFRS) in a given year, zero otherwise	Compustat Global and North America
$Patent\ count_{i,t}$	The natural logarithm of one plus the firm level count of triadic patents that are eventually granted; dated by application date (winsorized at the 1% level)	Worldwide Patent Statistical Database (PATSTAT)
$Patent\ citations_{i,t}$	The natural logarithm of one plus the count of future citations to patents that are eventually granted; dated by application date on the granted patent (winsorized at the 1% level)	Worldwide Patent Statistical Database (PATSTAT)
$\Delta MV_{i,t}/MV_{i,t-1}$	The percentage change in the market value of equity for firm i between year t and $t-1$ (winsorized at the 1% level)	Compustat Global and North America, Security Daily File
$R\&D_{i,t}/MV_{i,t-1}$	The ratio of R&D expenditures to the lagged market value of equity (winsorized at the 1% level)	Compustat Global and North America, Security Daily File
$Pollution\ taxes_{c,t}$	Taxes and charges directly applied to the pollution of sulfur oxides (SO _x). It is based on tax rate in Euros per tonne pollution by country and year. Categorized between 0 to 6, indicating low to high taxation levels	Environmental Stringency Index, PINE Databases (OECD)
$Pollution\ tax\ change_{c,t}$	Indicator variable equal to zero in the years before an increase in Pollution taxes, and one thereafter. If there are two changes in Pollution taxes, Pollution tax change starts at zero, increases to one after the first change in pollution taxes, and increases to two after the second change	Authors' calculations based on $Pollution\ taxes_{c,t}$
$SO_x\ emission_j$	Pounds of sulfur oxides (SO _x) per unit of output in each three-digit SIC industry in the U.S. manufacturing sector in 1987	Levinson 2009

- This table is the data dictionary of the paper.
- It clarifies that the main dependent variable is log R&D, the patent variable is $\log(1+\text{triadic patents})$, and the pollution-tax variable is an OECD categorical measure from 0 to 6.
- It also shows the exact scaling of the market-value specification and the origin of the pollution-intensity variable from Levinson (2009).

5.3 Table 2: Summary statistics

Read:

- Average log R&D is 1.038, while mean pollution tax is 0.959 with standard deviation 1.898.
- The typical firm has low average patenting and large variation in market-value changes.
- This table makes clear the tax variation is meaningful but concentrated, and that the firm-level outcomes are highly dispersed.

5.4 Table 3: Pollution intensity and R&D in selected industries

Read:

- The top-polluting industries are extremely dirty: cement, petroleum-and-coal products, inorganic chemicals, concrete/gypsum, steel, and smelting.

Table 2
Summary statistics

	Obs.	Mean	Median	SD
$R\&D_{i,t}$	33,545	1.038	0.325	1.428
$Sales_{i,t}$	33,545	2.778	2.297	2.718
$Cash-flow-to-assets_{i,t}$	33,343	0.076	0.084	0.154
$Sales\ growth_{i,t}$	33,343	0.042	0.035	0.306
$Cash-holdings-to-assets_{i,t}$	33,343	0.209	0.132	0.289
$Total\ debt-to-assets_{i,t}$	33,343	0.218	0.189	0.190
$IFRS_{i,t}$	33,343	0.196	0.000	0.397
$Patent\ count_{i,t}$	33,545	0.442	0.000	0.948
$Patent\ citations_{i,t}$	33,545	2.788	2.639	2.520
$\Delta MV_{i,t}/MV_{i,t-1}$	30,153	0.155	0.011	0.696
$R\&D_{i,t}/MV_{i,t-1}$	30,153	0.029	0.001	0.069
$Pollution\ taxes_{c,t}$	414	0.959	0.000	1.898
$Pollution\ tax\ change_{c,t}$	414	0.478	0.000	0.738
$SO_x\ emission_j$	107	9.216	1.086	21.603

This table reports the summary statistics for the key variables included in this study. The firm-level data are from Compustat Global and North America and consist of R&D reporting firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. We convert all dollar values to constant US\$ (2000) using the local currency unit conversion reported in Compustat Global and the annual U.S. GDP deflator from World Bank Development Indicators. Table 1 defines the variables in detail.

- The least-polluting industries are laboratory instruments, commercial printing, machinery, medical/optical goods, computers, and toys.
- The top 10 industries average 66.7 pounds of SO_x per unit of output, versus only 0.067 for the least-polluting 10, which validates the exposure design.

5.5 Table 4: Baseline relation between pollution taxes and R&D

Read:

- The baseline coefficient on pollution taxes is positive and highly significant: 0.062 without controls and 0.055 with controls.
- When tax changes are decomposed in event time, there is no lead effect before the tax rise.
- The post-change effect is about 0.136–0.156, which means R&D rises only after policy changes, not before them.

Table 4
Pollution taxes and R&D investment

	(1)	(2)	(3)	(4)	(5)	(6)
$Pollution\ taxes_{c,t-1}$	0.062 (0.020)***	0.055 (0.018)***				
$Sales_{i,t}$	0.312 (0.036)***	0.353 (0.038)***	0.311 (0.035)***	0.352 (0.037)***	0.311 (0.035)***	0.352 (0.037)***
$Pollution\ tax\ change_{c,t}$			0.149 (0.053)**	0.133 (0.052)**		
$Pollution\ tax\ change_{c,t}^{(-1)}$					0.006 (0.011)	−0.007 (0.012)
$Pollution\ tax\ change_{c,t}^{(0)}$					0.038 (0.025)	0.036 (0.026)
$Pollution\ tax\ change_{c,t}^{(\geq 1)}$					0.156 (0.056)**	0.136 (0.053)**
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm control set	No	Yes	No	Yes	No	Yes
Observations	33,545	33,343	33,545	33,343	33,545	33,343
Adjusted R^2	.954	.957	.955	.957	.955	.957

This table reports the OLS estimates for Equation (1). $R\&D_{i,t}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. All regressions include firm and year fixed effects. Columns 2, 4, and 6 also include the following firm-level control variables: *Cash-flow-to-assets*, *Sales growth*, *Cash-holdings-to-assets*, *Total debt-to-assets*, and *IFRS*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3
Pollution intensity and R&D investment in selected industries

SIC	Industry	SO _x emissions	R&D	# obs.
<i>A. Top-10 most-polluting industries</i>				
324	Cement, hydraulic	140.330	1.097	92
299	Misc. products of petroleum and coal	119.471	1.343	68
281	Industrial inorganic chemicals	82.974	0.715	467
327	Concrete, gypsum, and plaster products	71.692	0.504	296
331	Steel works, blast furnaces, mills	53.392	0.681	828
333	Primary smelting and refining	52.404	0.904	242
329	Abrasive, asbestos, and misc.	42.567	0.742	289
204	Grain mill products	36.625	0.439	223
287	Agricultural chemicals	34.352	0.754	287
286	Industrial organic chemicals	32.929	0.633	316
	Mean 10 most polluting	66.701	0.781	311
	Median 10 most polluting	52.898	0.729	288
<i>B. Ten least-polluting industries</i>				
382	Laboratory apparatus and instruments	0.141	0.926	1,309
275	Commercial printing	0.112	0.911	186
341	Metal cans and shipping containers	0.112	1.277	85
355	Special industry machinery, ex. metal	0.090	0.891	1,370
384	Photographic, medical, and optical goods	0.087	1.102	1,394
357	Computer and office equipment	0.085	1.054	1,016
205	Bakery products	0.024	0.143	111
394	Dolls, toys, games, and sporting	0.009	0.753	395
365	Household audio and video equipment	0.006	1.257	320
345	Screw machine products, bolts, nuts, etc.	0.001	0.248	132
	Mean 10 least polluting	0.067	0.856	632
	Median 10 least polluting	0.086	0.918	358

This table lists the 10 most (panel A) and 10 least (panel B) polluting industries (with more than 50 observations). *SO_x emissions* measures the total amount of pollution (in pounds) of sulfur oxides (SO_x) per unit of output in the United States based on data from Levinson (2009). R&D is the average *R&D* by industry. Table 1 defines the variables in detail.

Table 5
Pollution taxes and R&D investment: Differential effects in high-pollution industries

	(1)	(2)	(3)	(4)	(5)
$Pollution\ taxes_{c,t-1}$	0.000 (0.013)	−0.006 (0.011)			
$Sales_{i,t}$	0.311 (0.035)***	0.352 (0.038)***	0.259 (0.032)***	0.304 (0.036)***	0.297 (0.034)***
$Pollution\ taxes_{c,t-1} \times Q_2\ of\ SO_x\ polluters$	0.050 (0.016)***	0.042 (0.014)***	0.042 (0.016)**	0.033 (0.017)*	0.012 (0.036)
$Pollution\ taxes_{c,t-1} \times Q_3\ of\ SO_x\ polluters$	0.087 (0.042)*	0.089 (0.039)**	0.083 (0.022)***	0.086 (0.022)***	0.103 (0.033)***
$Pollution\ taxes_{c,t-1} \times Q_4\ of\ SO_x\ polluters$	0.097 (0.014)***	0.097 (0.011)***	0.103 (0.015)***	0.103 (0.013)***	0.092 (0.023)***
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	No	No
Firm control set	No	Yes	No	Yes	Yes
Country-year fixed effects	No	No	Yes	Yes	Yes
Industry-year fixed effects	No	No	No	No	Yes
Observations	33,545	33,343	33,529	33,327	33,003
Adjusted R^2	.955	.957	.959	.961	.961

This table reports the OLS estimates for Equation (1) augmented with interactions between country *Pollution taxes* and indicators for the industry pollution intensity quartiles. $R\&D_{i,t}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. All regressions include firm fixed effects; columns 1 and 2 (3–5) include year (country-year) fixed effects; and column 5 includes industry-year fixed effects. $Q_k\ of\ SO_x\ polluters$ is an indicator variable taking on the value one if the firm is located in the k th quartile in *SO_x emission*. Columns 2, 4, and 5 include the following firm-level control variables: *Cash-flow-to-assets*, *Sales growth*, *Cash-holdings-to-assets*, *Total debt-to-assets*, and *IFRS*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 5: Differential effects in high-pollution industries

Read:

- This is the paper's core causal table.
- The interaction of pollution taxes with the top pollution quartiles is strongly positive, around 0.08–0.10, and remains so with country-year and industry-year fixed effects.
- The uninteracted tax coefficient is near zero once differential exposure is emphasized, showing the action is in the dirty industries, not in the clean ones.

5.7 Figure 2: Robustness to alternative country-level mechanisms

Read:

- The figure shows that adding interactions for development, public environmental R&D, or the user cost of R&D barely moves the key interaction estimate.
- Hence the pollution-tax result is not a disguised effect of broad economic development or research-subsidy policy.
- The coefficient on the interaction with the top pollution quartile remains economically large and statistically meaningful in all cases.

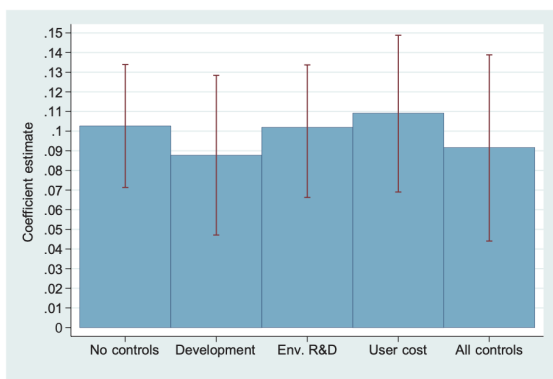


Figure 2
Robustness of the estimated relation between pollution taxes and R&D to alternative country-level mechanisms

This figure summarizes how adding a series of alternative country-level control variables to the augmented version of Equation (1) affects the coefficient for $Pollution\ taxes \times Q_4\ of\ SO_x\ polluters$ with R&D as dependent variable. The additional country-level variables are GDP per capita (*Development*), public environmental R&D to GDP (*Env. R&D*), and the user cost of R&D (*User cost*). The additional country-level variables are interacted with $Q_4\ of\ SO_x\ polluters$ and added to the regression alongside the firm and country-year fixed effects. Table 1 defines the variables in detail. Standard errors are clustered at the country level. The columns in the figure indicate the coefficient estimate on $Pollution\ taxes \times Q_4\ of\ SO_x\ polluters$, while the bands represent 95% confidence intervals.

Table 6
Pollution taxes and new invention

	(1)	(2)	(3)	(4)	(5)	(6)
	All technology classes			Air pollution abatement technologies		
$Pollution\ taxes_{c,t-1}$	-0.028 (0.031)	-0.008 (0.017)	-0.007 (0.012)	0.076 (0.024)***	0.090 (0.031)***	0.087 (0.028)***
$Stock\ of\ triadic\ patents_{i,t-1}$		0.437 (0.018)***	0.437 (0.018)***		0.311 (0.057)***	0.311 (0.057)***
$Pollution\ taxes_{c,t-1} \times Q_2\ of\ SO_x\ polluters$			0.002 (0.018)			-0.016 (0.018)
$Pollution\ taxes_{c,t-1} \times Q_3\ of\ SO_x\ polluters$			0.025 (0.022)			0.003 (0.040)
$Pollution\ taxes_{c,t-1} \times Q_4\ of\ SO_x\ polluters$			-0.016 (0.011)			0.010 (0.014)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,442	33,442	33,442	33,442	33,442	33,442
Adjusted R^2	.802	.846	.846	.198	.198	.198

This table reports the OLS estimates for Equation (1) (columns 1–2 and 4–5), and Equation (1) augmented with interactions between country $Pollution\ taxes$ and the indicators for the industry pollution intensity quartiles (columns 3 and 6). $Patent\ count_{i,t}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. Patents from technology classes classified as air pollution abatement technologies (based on Hascic and Migotto 2015; reported in Table B1) are reported in columns 4–6. $Q_k\ of\ SO_x\ polluters$ is an indicator variable taking on the value one if the firm is located in the k th quartile in $SO_x\ emission$. All regressions include firm and year fixed effects, $Sales$ and the following country-level control variables: *Public environmental R&D to GDP* and *GDP per capita*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.8 Table 6: Pollution taxes and new invention

Read:

- Total patenting does not increase with pollution taxes; the high-pollution firms that raise R&D do not disproportionately raise total patent counts.
- Clean air-pollution-abatement patents do rise in the full sample, with coefficients around 0.076–0.090.

- But the interaction of taxes with pollution intensity is near zero in the clean-patent regressions, which is the central fact distinguishing R&D from patent output.

5.9 Table 7: Pollution taxes and the marginal value of R&D

Read:

- In high-pollution industries, the interaction between pollution taxes and R&D/MV is strongly positive (1.431 or 0.685 depending on controls).
- In low-pollution industries, the corresponding interaction is negative and insignificant.
- The market therefore assigns higher value to marginal R&D when dirty firms face higher emissions taxes.

Table 7
Pollution taxes and the marginal value of R&D

	(1)	(2)	(3)	(4)
	High	Industry pollution intensity: High	High	Low
$R\&D_{i,t}/MV_{i,t-1}$	1.558 (0.322)***	0.749 (0.427)*	1.132 (0.353)***	1.562 (0.270)***
$Pollution\ taxes_{c,t} \times R\&D_{i,t}/MV_{i,t-1}$		1.431 (0.306)***	0.685 (0.220)***	-0.451 (0.554)
Firm fixed effects	Yes	Yes	Yes	Yes
Firm control set	Yes	Yes	Yes	Yes
Country year fixed effects	Yes	Yes	Yes	Yes
Firm control set x <i>Pollution taxes</i>	No	No	Yes	Yes
Observations	14,439	14,439	14,439	15,576
Adjusted R^2	.374	.378	.388	.370

This table reports the OLS estimates for Equation (2). $\Delta MV_{i,t}/MV_{i,t-1}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 1999. High (low) polluting industries in columns 1–3 (4) are industries above (below) the median in SO_x emission. All regressions include firm and country-year fixed effects. All columns include the following firm control variables (all scaled by lagged market value of equity): net assets, annual change in net income, dividends, cash holdings, net financing (net stock plus debt issuance), and total debt. Regressions in columns 3 and 4 include all firm control variables interacted with *Pollution taxes*. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Pollution taxes and R&D in low invention sectors

	(1)	(2)	(3)	(4)	(5)	(6)
	Low invention industry sort based on:					
	All triadic patent stock		Air pollution abatement patents			
$Pollution\ taxes_{c,t-1}$	0.061 (0.016)***	0.010 (0.022)	0.059 (0.017)***	0.019 (0.019)		
$Sales_{i,t}$	0.329 (0.026)***	0.329 (0.026)***	0.277 (0.025)***	0.333 (0.034)***	0.332 (0.033)***	0.276 (0.027)***
$Pollution\ taxes_{c,t-1} \times$ <i>Above-median SO_x polluters</i>		0.078 (0.014)***	0.082 (0.012)***		0.068 (0.015)***	0.076 (0.016)***
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	Yes	Yes	No
Firm control set	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Observations	16,106	16,106	16,069	25,872	25,872	25,856
R^2	.945	.945	.951	.953	.953	.958

This table reports the OLS estimates for Equation (1) (columns 1 and 4) and Equation (1) augmented with interactions between country *Pollution taxes* and the indicator for the industry pollution intensity variable (columns 2–3 and 5–6). $R\&D_{i,t}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 1999. Columns 1–3 (4–6) define low invention industries as the firms located in industries below the median in the stock of triadic patents (define low invention industries in air pollution abatement patents as not belonging to any of the high air pollution abatement patenting industries (which together make up about 80% of all patents in air pollution abatement patents during our sample period): Industrial inorganic chemicals (281), Motor vehicles and equipment (371), Pottery and related products (326), Computer and office equipment (357), General industrial machinery equipment (356), Flat glass (321), Refrigeration and service industry machinery (358), Industrial organic chemicals (286), and Fabricated rubber products, n.e.c. (306)). *Above-median SO_x polluters* is an indicator variable taking on the value one (zero) if the industry is above (below) the median in SO_x emission. All regressions include firm fixed effects, and columns 1–2 and 4–5 (3 and 6) include year (country-year) fixed effects. All columns include the following firm control variables: *Cash-flow-to-assets*, *Sales growth*, *Cash-holdings-to-assets*, *Total debt-to-assets*, and *IFRS*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.10 Table 8: R&D in low-invention sectors

Read:

- The paper first isolates sectors with low patent stocks or sectors outside the few industries generating most air-pollution-abatement patents.
- Even in these low-invention environments, the interaction of pollution taxes with above-median pollution intensity is strongly positive (around 0.068–0.082).
- This is strong evidence against the idea that the main response is patent-oriented invention.

5.11 Table 9: Process R&D

Read:

- In high-process-R&D sectors, pollution taxes have a large differential effect on dirty firms' R&D: the interaction is about 0.269–0.313.
- In low-process-R&D sectors, the same interaction is essentially zero.

- This strongly supports the interpretation that taxes induce process-oriented, absorptive-capacity investment.

Table 9
Pollution taxes and process R&D

	(1)	(2)	(3)	(4)	(5)	(6)
		High process R&D sectors			Low process R&D sectors	
<i>Pollution taxes</i> _{<i>c,t-1</i>}	0.097	-0.167		0.019	0.019	
	(0.019)***	(0.112)		(0.024)	(0.019)	
<i>Sales</i> _{<i>i,t</i>}	0.306	0.306	0.215	0.411	0.381	0.381
	(0.040)***	(0.041)***	(0.025)***	(0.050)***	(0.055)***	(0.055)***
<i>Pollution taxes</i> _{<i>c,t-1</i>} ^x	0.269	0.313		-0.001	-0.019	
<i>Above-median SO_x polluters</i>		(0.106)**	(0.090)***	(0.098)	(0.075)	
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	Yes	Yes	No
Firm control set	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Observations	12,060	12,060	12,037	18,328	18,328	18,306
Adjusted R ²	.952	.952	.960	.960	.960	.964

This table reports the OLS estimates for Equation (1) (columns 1 and 4) and Equation (1) augmented with interactions between country *Pollution taxes* and the indicator for the industry pollution intensity variable (columns 2–3 and 5–6). *R&D_{i,t}* is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. Columns 1–3 (4–6) define high (low) process R&D sectors as the firms located in industries above (below) the median in process R&D as measured in Cohen and Klepper (1996). *Above-median SO_x polluters* is an indicator variable taking on the value one (zero) if the industry is above (below) the median in *SO_x emission*. All regressions include firm fixed effects, and columns 1–2 and 4–5 (3 and 6) include year (country-year) fixed effects. All columns include the following firm control variables: *Cash-flow-to-assets*, *Sales growth*, *Cash-holdings-to-assets*, *Total debt-to-assets*, and *IFRS*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. **p* < .1; ***p* < .05; ****p* < .01.

Table 10
Pollution taxes and R&D: Splits based on knowledge spillovers

	(1)	(2)	(3)	(4)	(5)	(6)
		High spillovers			Low spillovers	
<i>Pollution taxes</i> _{<i>c,t-1</i>}	0.068	0.016		0.034	0.025	
	(0.017)***	(0.016)		(0.024)	(0.037)	
<i>Sales</i> _{<i>i,t</i>}	0.335	0.334	0.269	0.399	0.399	0.362
	(0.030)***	(0.029)***	(0.024)***	(0.063)***	(0.063)***	(0.068)***
<i>Pollution taxes</i> _{<i>c,t-1</i>} ^x	0.090	0.094		0.019	0.011	
<i>Above-median SO_x polluters</i>		(0.007)***	(0.014)***	(0.061)	(0.075)	
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	Yes	Yes	No
Firm control set	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Observations	15,348	15,348	15,312	13,204	13,204	13,181
Adjusted R ²	.951	.951	.957	.965	.965	.969

This table reports the OLS estimates for Equation (1) (columns 1 and 4) and Equation (1) augmented with interactions between country *Pollution taxes* and the indicator for the industry pollution intensity variable (columns 2–3 and 5–6). *R&D_{i,t}* is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 3999. Columns 1–3 (4–6) define high (low) spillover sectors as the industries below (above) the median in appropriability as measured in Cohen, Nelson, and Walsh (2000). *Above-median SO_x polluters* is an indicator variable taking on the value one (zero) if the industry is above (below) the median in *SO_x emission*. All regressions include firm fixed effects, and columns 1–2 and 4–5 (3 and 6) include year (country-year) fixed effects. All columns include the following firm control variables: *Cash-flow-to-assets*, *Sales growth*, *Cash-holdings-to-assets*, *Total debt-to-assets*, and *IFRS*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. **p* < .1; ***p* < .05; ****p* < .01.

5.12 Table 10: High-spillover versus low-spillover sectors

Read:

- In high-spillover / low-appropriability sectors, the interaction of taxes with above-median pollution intensity is about 0.090–0.094 and highly significant.
- In low-spillover / high-appropriability sectors, the interaction is small and insignificant.
- This is the cleanest “second face” result: the R&D response is strongest where external knowledge is easier to absorb and where pure invention incentives are weaker.

5.13 Table 11: The market value of the second face of R&D

Read:

- The market-value regression is repeated within the subsets where the absorptive-capacity interpretation is strongest.
- In five of six specifications, the interaction between pollution taxes and R&D remains positive and significant, typically between 0.651 and 1.411.
- This implies that the second-face R&D response is value-enhancing, not merely an accounting or managerial artifact.

5.14 Table 12: Do any high-pollution firms innovate?

Read:

- Among high-pollution firms, both the prior-clean-innovation and no-prior-clean-innovation groups raise R&D by similar amounts (0.092 and 0.093).

Table 11

Pollution taxes and the market value of the second face of R&D

	(1)	(2)	(3)	(4)	(5)	(6)
	Low invention sectors		High process R&D sectors		High spillover sectors	
$R\&D_{i,t}/MV_{i,t-1}$	0.667 (0.373)*	0.992 (0.399)**	0.856 (0.395)**	1.314 (0.415)***	0.664 (0.339)*	1.017 (0.338)***
$Pollution\ taxes_{c,t} \times R\&D_{i,t}/MV_{i,t-1}$	1.375 (0.297)***	0.717 (0.228)***	1.321 (0.298)***	0.352 (0.287)	1.411 (0.300)***	0.651 (0.234)**
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm control set	Yes	Yes	Yes	Yes	Yes	Yes
Country year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm control set x <i>Pollution taxes</i>	No	Yes	No	Yes	No	Yes
Observations	7,913	7,913	9,541	9,541	8,271	8,271
Adjusted R^2	.378	.385	.384	.398	.396	.406

This table reports the OLS estimates for Equation (2). $\Delta MV_{i,t}/MV_{i,t-1}$ is the dependent variable. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 1999. All firms are located in high polluting sectors, defined as industries above the median in SO_x emission. Columns 1 and 2 define low invention sectors as the firms located in sectors below the median in the stock of triadic patents. Columns 3 and 4 define high process R&D sectors as the industries above the median in process R&D as measured in Cohen and Klepper (1996). Columns 5 and 6 define high spillover sectors as the industries below the median in appropriability as measured in Cohen, Nelson, and Walsh (2000). All regressions include firm and country-year fixed effects and the following firm control variables (all scaled by lagged market value of equity): net assets, annual change in net income, dividends, cash holdings, net financing (net stock plus debt issuance), and total debt. Regressions in even-numbered columns include all firm control variables interacted with *Pollution taxes*. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 12

Pollution taxes and clean invention in high-pollution firms

	(1)	(2)	(3)	(4)
	Industry air pollution abatement patenting in 1990:			
	High		Low	
	R&D		Patent count	
$Pollution\ taxes_{c,t-1}$	0.092 (0.068)	0.093 (0.020)***	0.445 (0.208)**	0.005 (0.003)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	5,593	10,627	5,581	10,596
Adjusted R^2	.965	.944	.257	.000

This table reports the OLS estimates for Equation (1). $R\&D_{i,t}$ is the dependent variable in columns 1 and 2 and *Patent count* _{i,t} is the dependent variable in columns 3 and 4. The firm-level data are from Compustat Global and North America and consist of non-U.S. firms with at least three nonmissing R&D observations during 1990–2012 and a primary SIC industry classification between 2000 and 1999. Only firms from industries above the median in SO_x emission are included. Patent count based on technology classes classified as air pollution abatement technologies (based on Hascic and Migotto 2015; reported in Table B1) are reported in columns 3 and 4. All regressions include firm and year fixed effects and *Sales*. Regressions in columns 3 and 4 include *Stock of triadic patents* _{$i,t-1$} and the following country-level control variables: *Public environmental R&D to GDP* and *GDP per capita*. Table 1 defines the variables in detail. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

- But only the subset with a prior history of innovation in air-pollution-abatement technologies increases clean patenting (0.445).
- This is a path-dependence result: pollution taxes induce new invention only where firms already have innovative capability in that technological niche.

6 Short summary

- **Conclusion.** Emissions taxes induce dirty firms to spend more on R&D, but the extra R&D usually does not show up as more overall patenting.
- **Mechanism.** The response is best understood as investment in absorptive capacity and process improvement rather than standard new-product invention.
- **Evidence.** The effect is strongest in dirty industries, in low-invention sectors, in high-process R&D sectors, and in high-spillover sectors; and the market values this R&D more highly under higher taxes.
- **Nuance.** A subset of already-innovative dirty firms *does* increase clean pollution-abatement patents, which fits path dependence in technical change.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Higher taxes on dirty emissions raise firm-level R&D spending, especially in the dirtiest industries.
- **What the paper rejects.** The increase in R&D is not mainly a broad increase in patentable invention by the high-pollution firms that are most treated by the tax.
- **What the paper supports instead.** Pollution taxes raise the payoff to technological capability that helps firms adopt cleaner processes and assimilate outside knowledge.
- **Value implication.** The market values a marginal dollar of R&D more highly in polluting firms facing higher taxes, including in the exact subsamples where the “second face” interpretation is

strongest.

7.2 Economic intuition

- Dirty firms face higher operating costs when emissions are taxed. This makes cleaner production methods more attractive at the margin.
- For many incumbent firms, switching production methods does not first require a patentable invention; it requires the capability to understand, acquire, and implement external technical knowledge.
- In that sense, emissions taxes increase the marginal return to absorptive-capacity R&D even if patent counts do not rise.

7.3 Takeaway

This paper is important because it separates R&D from patenting in a context where the two should not be treated as interchangeable. Environmental policy can redirect technical change not only by inducing clean inventions, but also by pushing dirty firms to invest in the organizational and technological capabilities required to use cleaner production methods. The big lesson is that the “second face” of R&D—absorbing and implementing outside knowledge—is a first-order channel through which emissions taxes affect firms.